0.1 $\epsilon - \delta$ Limits, Uniform Continuity, and the Bolzano-Weierstrass Theorem

Building upon the topological framework of R, we rigorously define the limiting behavior of functions and sequences, transitioning from local properties to global guarantees on compact domains.

Let $E \subset R$, and let c be a limit point of E. A function $f : E \to R$ has a **limit** $L \in R$ as x approaches c, denoted $\lim_{x\to c} f(x) = L$, if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in E$,

$$0 < |x - c| < \delta |f(x) - L| < \epsilon$$

From this local definition, we define continuity: a function f is continuous at $c \in E$ if $\lim_{x\to c} f(x) = f(c)$. If this holds for all points in E, f is continuous on E. While continuity is a local property (the choice of δ may depend on both ϵ and c), uniform continuity establishes a global bound independent of c.

A function $f : E \to R$ is **uniformly continuous** on E if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in E$,

$$|x - y| < \delta |f(x) - f(y)| < \epsilon$$

[Heine-Cantor Theorem] If $K \subset R$ is compact and $f : K \to R$ is continuous, then f is uniformly continuous on K.

Let $\epsilon > 0$ be given. Because f is continuous on K, for each $p \in K$, there exists a $\delta_p > 0$ such that for all $x \in K$,

$$|x - p| < \delta_p |f(x) - f(p)| < \frac{\epsilon}{2}$$

Consider the collection of open neighborhoods $V_p = (p - \frac{\delta_p}{2}, p + \frac{\delta_p}{2})$ for all $p \in K$. The collection $\{V_p\}_{p\in K}$ forms an open cover of K. Because K is compact, there exists a finite subcover corresponding to points $p_1, p_2, \ldots, p_n \in K$.

Define $\delta = \frac{1}{2} \min\{\delta_{p_1}, \delta_{p_2}, \ldots, \delta_{p_n}\}$. Since the set of radii is finite and each $\delta_{p_i} > 0$, we have $\delta > 0$. Let $x, y \in K$ such that $|x - y| < \delta$. Because the finite collection $\{V_{p_i}\}_{i=1}^n$ covers K, there is some integer $k \in \{1, \ldots, n\}$ such that $x \in V_{p_k}$. This implies:

$$|x - p_k| < \frac{\delta_{p_k}}{2}$$

By the triangle inequality, we bound the distance between y and p_k:

$$|y - p_k| \leq |y - x| + |x - p_k| < \delta + \frac{\delta_{p_k}}{2}$$

Since $\delta \leq \frac{\delta_{p_k}}{2}$, we deduce $|y - p_k| < \delta_{p_k}$. We evaluate the difference in functional values using the continuity of f at p_k. Both x and y are strictly within δ_{p_k} of p_k, yielding:

$$|f(x) - f(y)| \leq |f(x) - f(p_k)| + |f(p_k) - f(y)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus, f is uniformly continuous on K.

We conclude this chapter with a fundamental property of sequences in bounded domains, relying heavily on the completeness axiom.

[Bolzano-Weierstrass Theorem] Every bounded sequence in R possesses a convergent subsequence.

Let $(x_n)_{n=1}^{\infty}$ be a bounded sequence in R. There exists a closed interval $I_0 = [a_0, b_0]$ such that $x_n \in I_0$ for all $n \in N$. We proceed by constructing a nested sequence of closed intervals.

Bisect I_0 into two closed subintervals: $[a_0, \frac{a_0+b_0}{2}]$ and $[\frac{a_0+b_0}{2}, b_0]$. Since I_0 contains infinitely many terms of the sequence (x_n), at least one of these two subintervals must contain infinitely many terms. Let $I_1 = [a_1, b_1]$ denote this subinterval. We select an index n_1 such that $x_{n_1} \in I_1$.

Assume inductively that we have constructed a closed interval $I_k = [a_k, b_k]$ containing infinitely many terms of (x_n), with length $\frac{b_0-a_0}{2^k}$, and we have chosen an index $n_k > n_{k-1}$ such that $x_{n_k} \in I_k$. Bisect I_k into two equal halves. At least one half contains infinitely many terms of (x_n); call it $I_{k+1} = [a_{k+1}, b_{k+1}]$. Choose an index $n_{k+1} > n_k$ such that $x_{n_{k+1}} \in I_{k+1}$.

This process recursively generates a sequence of nested closed intervals $I_0 \supset I_1 \supset I_2 \supset \ldots \supset I_k \supset \ldots$ and a subsequence $(x_{n_k})_{k=1}^{\infty}$ where $x_{n_k} \in I_k$ for all k. The sequences of endpoints (a_k) and (b_k) are monotonic: (a_k) is non-decreasing and bounded above by b_0, while (b_k) is non-increasing and bounded below by a_0. By the Monotone Convergence Theorem (derived from the Least Upper Bound property), both converge. Since the interval length is $b_k - a_k = \frac{b_0-a_0}{2^k}$, we have $\lim_{k\to\infty}(b_k - a_k) = 0$, proving they converge to the identical limit $L \in R$.

Because $a_k \leq x_{n_k} \leq b_k$ for all k, the Squeeze Theorem forces $\lim_{k\to\infty} x_{n_k} = L$. Thus, (x_{n_k}) converges.